

Part: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3402**

Title: **STRENGTH OF COMPOSITE LAMINATES**

Revision: **0.1**

Date: **19/01/2021**

1 INTRODUCTION

This is an introductory chapter listing the applicable stress analysis method necessary to check the strength of composite laminates under static loads in stable structural details (buckling stability not treated here).

2 STRENGTH OF COMPOSITE LAMINATES

2.1 PLAIN STRENGTH

This applies to composite details with pristine/undamaged material (beyond acceptable quality tolerances), without any feature which could introduce stress concentrations. The stress method for this case is discussed in detail in chapter 3403 “PLAIN STRENGTH OF COMPOSITE LAMINATES”.

2.2 STRENGTH OF NOTCHED LAMINATES

This applies to composite details including some kind of stress raiser, like holes or cracks. The stress method for this case is discussed in detail in chapter 3404 “STRENGTH IN NOTCHED LAMINATES”.

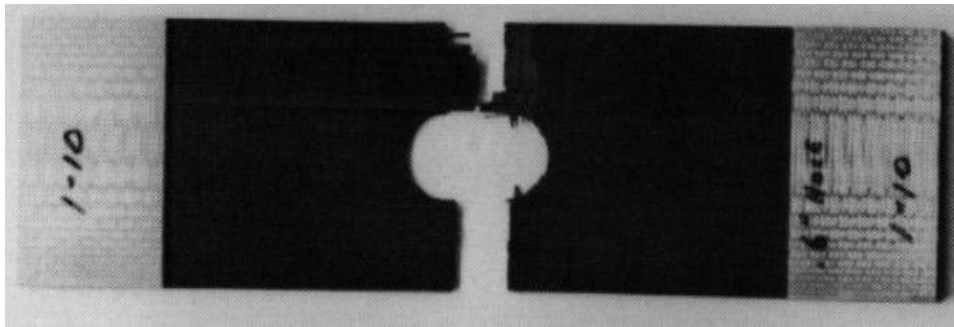


Figure 1. Open-hole test coupon after failure

2.3 RESIDUAL STRENGTH AFTER LOW VELOCITY IMPACT

Standard certification regulations demand primary composite structures to have ultimate load capability even when included damages caused by low velocity impacts in the threshold of detectability. Find a more exhaustive discussion about damage tolerance considerations in composite materials in chapter 4910 “FATIGUE ASSESSMENT OF COMPOSITE PARTS”. This section deals with stress methods applicable to assess this capability up to ultimate loads level.

2.3.1 DT-Strength after transverse impact in panel

The stress method for this case is discussed in detail in chapter 3405 “RESIDUAL STRENGTH AFTER TRANSVERSE BVID IMPACT”.

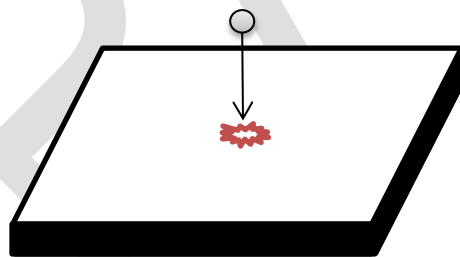


Figure 2. Transverse impact at panel centre

The approximation for the design values in chapter 1421 “MODEL FOR RESIDUAL STRENGTH AFTER IMPACT IN COMPOSITE LAMINATES”. That approximation model is derived from tests in “standard” CAI coupons. At this point, it is considered relevant to remind this extract taken Airbus DS extended testing method (Ref. 2):

“3.2 IMPACTS IN PARTS VS IMPACTS IN COUPONS

Normally it will be very difficult, or even impossible, to get comparable support conditions for the impact tests in the real part and in the coupon. Then, the observed damage result, typically characterized in terms of dent depth and delamination area, will not be the same in the coupon and in the real part, even the same impactor and kinetic energy were used to introduce the impact damage.

Different techniques were used in the past to get consistent pairs of energy-damage in coupons and real parts. General recommendation is that this kind of artefacts (e.g. using an impactor of higher diameter in coupons tests to retard penetration and increase the projected delaminated area) shall be avoided.

General suggested approach:

- ✓ *Design Values obtained using this test method shall be applicable to composite plates/shells areas of, at least, a characteristic dimension of 75mm (between stiffeners/supports, edges,...). It is generally accepted that DV's obtained from tests at coupon level shall be conservative for the panels.*
- ✓ *For zones adjacent to free edges, or under a stiffener/support, or stiffeners flanges of width lower than that characteristic dimension of 75mm, dedicated detail tests considering all the features of the real part (e.g. stiffener and effective skin) shall be done instead."*

2.3.2 DT-Strength after transverse impact in straight edge

The same method and allowable strain values proposed for the general case of transverse impacts on panels shall be used.



Figure 3. Transverse impact at panel edge

The support of this design policy is that, for comparable detectability thresholds and impact energy limits the residual strength near a panel edge shall be greater than the standard CAI design allowable (derived from coupons 100mm wide supported along its perimeter), as:

- ✓ At panel edges a greater part of the impact energy will be released by elastic bending (instead of damaging the material)
- ✓ In addition to dent depths, delamination cracks at edges become damage indicators during inspections (increased detectability)
- ✓ For impacts near panel edges the characteristic size of the damage (D_{d-edge}) shall be lower than that the corresponding value in the coupon (D_{d-ref})

Damage severity at BVID threshold shall be higher in CAI coupons (ref. design values), their resistant net-section ($1-d/w$) and residual strength lower than those at impacted edges. Thus, sizing with standard design values for CAI shall be conservative for panels edges.

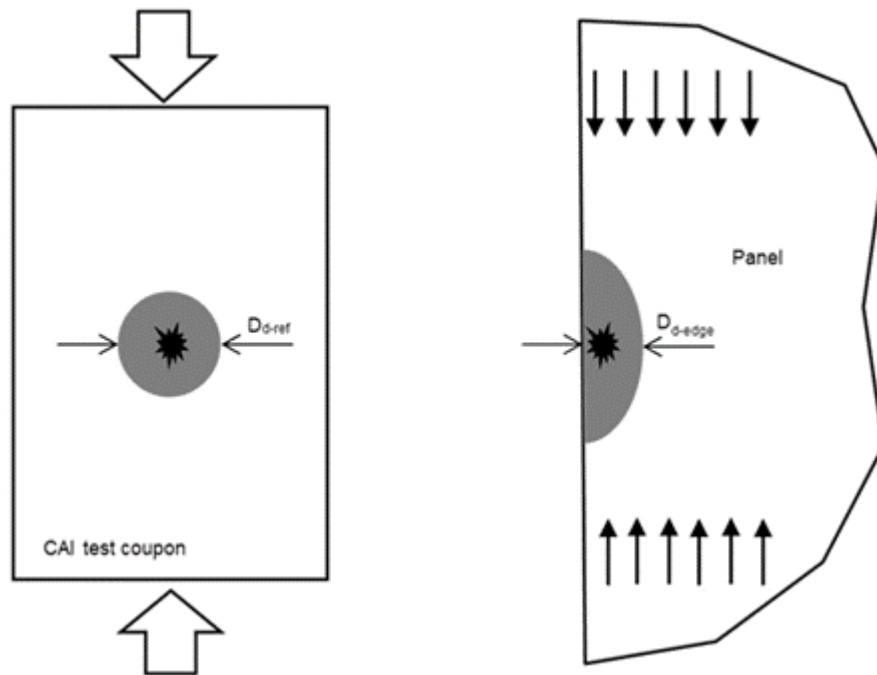


Figure 4. Transverse impact at centre of panel/coupon and at straight edge

Some substantiation of this? ... The memorandum being prepared for the Future Falcon?

2.3.3 DT-Strength after transverse impact in hole edge

A dedicated allowable strain for this case shall be compared with the peak tangential strain calculated along the edge of a lightening hole. In case no specific design value is available for the material and configuration, the allowable suggested, valid only for preliminary design, shall be twice the gross strength derived from standard CAI coupons (*).

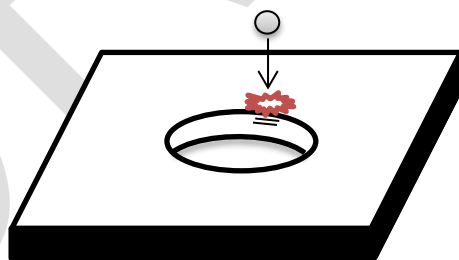


Figure 5. Transverse impact at hole edge

From a distance to the hole edge in which stress concentration effects are less relevant (**), the allowable design values proposed for the general case of transverse impacts on panels shall be used instead.

(*) The approach shall have to be validated by dedicated testing at some moment of the design/certification process

(**) Just a fraction of the hole radius and of the same order as the BVID size

2.3.4 DT-Strength after edge impact in panel

The approach for this case shall be similar for transverse impacts at straight panel edges.



Figure 6. Edge impact at panel

2.3.5 DT-Strength after edge impact in stringer

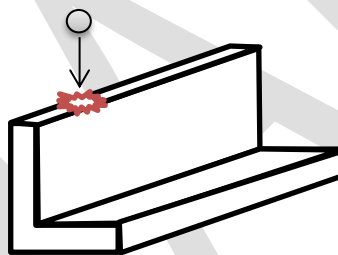
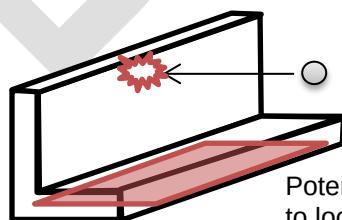


Figure 7. Edge impact in stringer

We do not have dedicated tests data... Recommend some kind of cut-off preliminary design value inspired in external sources (and standard CAI) pending validation by dedicated detail/part tests?

2.3.6 DT-Strength after transverse impact in stringer



Potential disbond of foot in addition to local damage at point of impact

Figure 8. Transverse impact in stringer

Local damage at impact location

We shall not have dedicated tests data for any stringer configuration... Recommend the use of the standard CAI (based in flat coupons 100mm wide) pending validation by dedicated detail/part tests?

Foot disbond

Recommend DFEM simulation considering a disbond of 100mm length (*) and a foot width from stringer web plane to foot edge (**)?

(*) Legacy detectability threshold set for Fan Cows omegas

(**) From observations in impact tests on panels, i.e. whole width disbanded in L stringers, but only half width disbanded in T stringers.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
A/C		Aircraft (style "Abbreviation")
CAI	MPa/με	Compression After Impact strength
TBC		To Be Confirmed

REFERENCES

	Document Reference	Issue	Title
Ref. 1	TAE-XX-XX-XXXXXX	X	This is the reference number one (style "Reference"). In this case, an especially long document title is used.
Ref. 2	TEA-T-SP-170011	Is. A	TEST METHOD FOR DETERMINATION OF THE STRENGTH AFTER IMPACT OF SOLID LAMINATES MADE OF FIBRE REINFORCED PLASTICS.

SOFTWARE

Program	Version	Description

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
0.1 19/01/2021	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEPAP-TL4)	G. Baños 19/01/2021	F. Glock DD/MM/2021

Table 3. Approval signatures table for last revision of this chapter